4WIDS-MAPF: Multiagent Pathfinding for Four-Wheel Independent Drive/Steering Robots

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***Abstract*—Modern autonomous systems face significant challenges in coordinating multiple agents within complex environments, especially under nonholonomic constraints and kinodynamic motion planning demands. In particular, four-wheel independent drive/steering (4WIDS) robots, with their multimodal locomotion and nonholonomic kinematics, introduce new complexity to multiagent pathfinding problems in continuous space. This study addresses the critical challenge of generating collision-free, dynamically feasible paths for multi-4WIDS robot systems. We propose a hierarchical multiagent path finding (MAPF) framework that explicitly tackles two core limitations in existing methods: 1) a lack of physically realistic interagent conflict modeling; and 2) the inability to account for motion mode switching costs in kinodynamic path planning. At the upper level, we introduce a continuous-space binary conflict tree for resolving space–time collisions using physical dimensions and motion feasibility. At the lower level, we implement a hybrid A* search over a motion primitive database that considers mode-switch penalties and kinematic constraints. To improve search performance, we introduce two novel mechanisms: 1) a focused search factor for directional exploration and 2) an adaptive heuristic weighting factor to balance optimality and computational speed. Extensive experiments, including benchmark comparisons, ablation studies, and sensitivity analysis, validate the proposed method's superior performance in interagent conflict resolution, real-time feasibility, and path quality.**

***Index Terms*—Enhanced conflict-based search (CBS), four-wheel independent drive/steering (4WIDS) robot, multiagent path finding (MAPF), suboptimal solution acceleration.**

I. Introduction

MULTIAGENT path finding (MAPF) constitutes a fundamental research challenge in IoT-enabled collaborative autonomous systems, serving as the computational backbone for smart industrial logistics, emergency response networks, and intelligent transportation ecosystems [1], [2]. While single-agent navigation represents a well-studied domain, the extension to multiagent systems introduces exponential complexity arising from spatio-temporal entanglement and combinatorial interdependencies [3], [4]. Among these challenges, nonholonomic kinematic constraints velocity-level restrictions that cannot be integrated into simple geometric relations, such as the pure rolling and no-slip conditions of wheeled robots pose a significant hurdle. Unlike holonomic agents, whose motion is solely determined by position and orientation (e.g., drones with omnidirectional movement), nonholonomic systems must navigate through differential constraints, limiting their instantaneous motion directions while enabling global mobility through strategic steering and translation [5], [6]. Effective MAPF solutions must thus balance collision-free coordination with compliance to these agent-specific dynamics, a dual requirement that remains particularly challenging for nonholonomic platforms.

Conventional MAPF methods use discrete spatial models with omnidirectional agents, ignoring physical dimensions and dynamic constraints. This abstraction gap becomes critically apparent when considering emerging robotic platforms like four-wheel independent drive/steering (4WIDS) robots. These advanced systems combine differential drive responsiveness with Ackermann steering (a steering geometry where the front wheels turn at different angles to maintain pure rolling during cornering, widely used in automobiles) through four independently controlled steering/driving motors, enabling multimodal locomotion, including lateral translation and zero-radius rotation. Such kinematic versatility makes 4WIDS robots particularly suitable for heavy duty operations in constrained environments, such as automated warehouses and industrial complexes [7], [8], [9]. However, their enhanced maneuverability introduces complex motion constraints that fundamentally challenge traditional MAPF assumptions, necessitating novel algorithmic frameworks that reconcile discrete coordination strategies with continuous motion planning.

This article bridges this critical gap by developing a specialized MAPF framework for 4WIDS robots in continuous workspaces (Fig. 1). Our methodology extends the principles of conflict-based search (CBS) through three key innovations that address the unique challenges posed by multimodal locomotion capabilities. Through systematic experimentation and theoretical analysis, we demonstrate that our approach achieves superior coordination efficiency while maintaining computational tractability. The principal scientific contributions of this work include

1) We develop a hierarchical planner that integrates high-level space–time physical conflict resolution and low-level motion primitive search. This design

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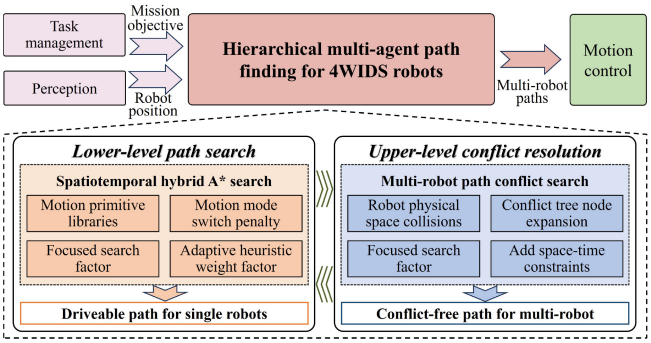


Fig. 1. Hierarchical MAPF framework for 4WIDS robots.

overcomes the limitations of grid-based CBS methods by enabling physically realistic coordination of nonholonomic, multimodal 4WIDS agents in continuous workspaces.

2) We introduce an adaptive heuristic search strategy that employs dominance rules and conflict prioritization to accelerate the search process, with sensitivity and ablation experiments confirming a balance between optimality and efficiency.
3) We conduct comprehensive verification through benchmark analysis and application scenarios testing, quantitatively demonstrating the superiority of our approach compared to advanced MAPF methods.

This article is organized as follows. Section II reviews related work. Section III introduces the preliminaries. Sections IV and V detail the proposed methodology and experimental results, respectively, and Section VI concludes this article.

II. Related Work

Current research on 4WIDS systems primarily focuses on motion control at the single-agent level, with particular emphasis on path-tracking performance, while relatively limited attention has been paid to multiagent coordination. Both 4WIDS vehicles and 4WIDS robots share similar actuation architectures and dynamic models, which often leads to a lack of distinction between them in existing studies. However, 4WIDS robots are generally deployed in more dynamic and constrained environments, where task flexibility, real-time decision-making, and collaborative control are critical. For instance, Wang et al. [10] developed an adaptive model predictive control (MPC) strategy that enhances 4WIDS robot path-tracking accuracy by estimating tire lateral stiffness via recursive least squares and integrating a torque distribution algorithm. Lin et al. [11] proposed a reinforcement learning-based linear MPC controller that compensates for unknown dynamics and optimizes the weight matrix, ensuring stability through Lyapunov analysis. Nguyen et al. [12] introduced an MPC and mixed-integer quadratic programming framework for 4WIDS vehicles, autonomously switching between parallel positive steering and symmetric negative steering to optimize tracking performance. Xu et al. [13] explored dynamic switching control for rescue vehicles with 4WIDS, leveraging B-spline curves and an improved nondominated sorting genetic algorithm to optimize switching trajectories, achieving a constant turning radius and minimizing energy consumption. Though these studies significantly enhance individual maneuverability through sophisticated mode-switching mechanisms, they fundamentally address single-robot scenarios, leaving multiagent coordination challenges unexamined.

MAPF has been extensively studied in robotics and AI, with early approaches extending single-agent motion planners, such as A*, cost expansion tree search, and conflict-based methods [14], [15]. Recent research has shifted toward conflict resolution and optimization-based strategies. For example, Kottinger et al. [16] introduced kinodynamic CBS, which operates in continuous space, incorporates motion constraints, and preserves probabilistic completeness. Wen et al. [17] proposed CL-CBS, which accounts for agent shapes and resolves conflicts via a body collision tree alongside a spatio-temporal A* planner with a sequential variant for efficiency. Ren et al. [18] developed multiobjective CBS to address search space expansion in MAPF. Moldagalieva et al. [19] integrated CBS with discontinuous-boundary A* to handle diverse robot dynamics in complex environments. Despite these advances, current approaches still exhibit two key limitations: 1) an oversimplification of kinematic constraints, particularly for nonholonomic systems; and 2) reduced solution speed and increased computational complexity as the number of robots and scene complexity rise, often necessitating extensive post-processing optimization.

In the optimization-based strategies domain, Yan and Li [20] proposed a three-level planner, PSB, that integrates agent dynamics to generate smooth velocity trajectories. Zang et al. [21] introduced a trajectory planning and motion optimization framework using spatio-temporal safety corridors, mixed-integer quadratic programming, and motion optimization to ensure safe and smooth trajectory generation for multiple automated guided vehicles in complex dynamic environments. Ma et al. [22] developed a decentralized swarm planning framework that leverages environmental topology and a search-based speed planning strategy to generate safe, smooth, and dynamically feasible trajectories in real time. While these methods offer accurate path optimization, they often rely on idealized assumptions. Additionally, they are computationally intensive, making real-time solutions challenging as the number of agents or environmental complexity increases.

In recent years, reinforcement learning has emerged as a promising approach for MAPF problems [23], [24], [25]. Yun et al. [26] combined large-neighborhood search with multiagent reinforcement learning (MARL) for efficient conflict resolution, using MARL in early stages and priority planning later to improve computational efficiency. Chung et al. [27] developed TFDueling, a deep reinforcement learning-based method for robot autonomy in target selection and path planning. Li et al. [28] integrated deep reinforcement learning, MPC, and graph neural networks to optimize path planning and task allocation in multirobot systems. While achieving notable success in specific domains, learning-based methods face inherent limitations in interpretability,

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convergence stability (especially in sparse-reward scenarios), and substantial computational resource requirements for policy training.

In this article, we address the challenges posed by physical dimensions collisions and the unique mobility of 4WIDS robots. We integrate high-level spatio-temporal conflict resolution with low-level robot motion primitive search to generate conflict-free trajectories for multiple robots. Additionally, we propose a suboptimal accelerated search strategy that balances scalability, solution speed, and quality in complex operational scenarios.

III. Preliminaries

A. Problem Definition

In a 2-D continuous shared space $\mathcal{W} \in \mathbb{R}$, a group of agents $\mathcal{A} = \{a_1, a_2, \ldots, a_m\}$ operates alongside known impassable regions $\mathcal{O}$, defining their free workspace $F = \mathcal{W} \setminus \mathcal{O}$. Each agent a_i at time t occupies state space ${s_i}^t$ with a collision footprint $\mathcal{R}({s_i}^t) \in \mathcal{W}$. For each agent in the group, unique start points $\mathcal{S} = \{{s_1}^0, {s_2}^0, \ldots, {s_m}^0\}$ and goals $G = \{g_1, g_2, \ldots, g_m\}$ are allocated, ensuring discrete paths $\mathcal{P}_i = \{{s_i}^0, {s_i}^1, \ldots, {s_i}^{t_T}\}$ are planned, adhering to

$$\mathcal{P}_i[0] = {s_i}^0 \quad \forall i \in \mathcal{A} \tag{1}$$

$$\mathcal{P}_i[t_T] = g_i \quad \forall i \in \mathcal{A} \tag{2}$$

$$R(\mathcal{P}_i[t]) \cap o = \varnothing \quad \forall i \in \mathcal{A}, 0 \le t \le t_T, o \in \mathcal{O} \tag{3}$$

$$\mathcal{R}(\mathcal{P}_i[t]) \cap \mathcal{R}(\mathcal{P}_j[t]) = \varnothing \quad \forall i, j \in \mathcal{A}, i \ne j, 0 \le t \le t_T. \tag{4}$$

Conditions 1 and 2 ensure that each agent adheres to its designated start and goal positions. Condition 3 requires agents to avoid collisions with obstacles. Condition 4 guarantees collision-free interactions among agents.

To evaluate the performance, metrics, such as runtime, average execution time, average travel distance, and total cost are defined to assess each planner's solution. Runtime refers to the duration required for the multiagent system to find paths. Average execution time measures the time taken for the multiagent system to complete tasks from the start to the target locations. Average travel distance represents the total distance traveled along the planned paths. Total cost encompasses the expenses incurred for moving agents from their current positions to the target locations. Due to the multimodal nature of their movements, total cost cannot be solely determined by distance. It should be noted that this study focuses solely on conflict-free multiagent path planning, assuming perfect trajectory tracking; controller design and performance are therefore outside its scope.

B. Robot Motion Model

As shown in Fig. 2, the 4WIDS robot studied in this article adopts Ackermann steering geometry, commonly used for car-like robots, as its fundamental kinematic model [29]. The robot's state is represented by ${s_i}^t = [x_i(t), y_i(t), \theta_i(t)]$. In addition to this model, lateral and rotational motion models are incorporated, along with the introduction of waiting nodes [30]. To facilitate path planning and search, the robot's

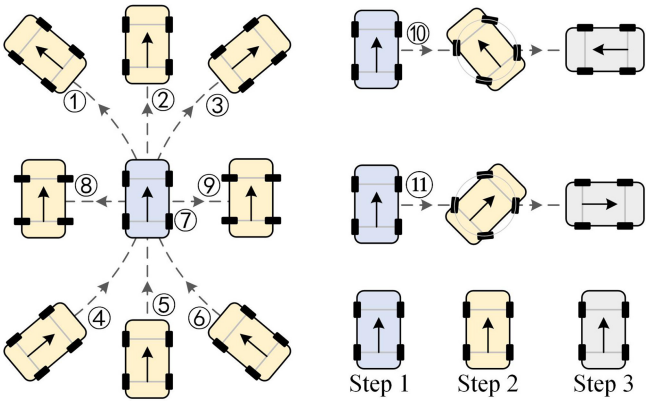


Fig. 2. 4WIDS robots different motion modes: Forward left, forward straight, forward right, backward left, backward straight, backward right, wait, lateral left, lateral right, rotate left, and rotate right.

multimodal capabilities are utilized. The robot's motion in the inertial frame is described by

$$\begin{bmatrix} \dot{x}_i(t) \\ \dot{y}_i(t) \\ \dot{\theta}_i(t) \end{bmatrix} = \begin{bmatrix} v_i(t)\cos(\theta_i(t)) \\ v_i(t)\sin(\theta_i(t)) \\ v_i(t)\frac{\tan(\beta_i(t))}{L_i} \end{bmatrix} \tag{5}$$

where $i \in \{1, 2, \ldots, m\}$, $t \in [0, t_T]$, the origin of the rigid-body coordinate system $(x_i(t), y_i(t))$ is located at the center of robot i's body, where $\theta_i(t)$ denotes the orientation angle of robot i and L_i represents robot i's wheelbase. The control inputs $[v_i(t), \beta_i(t)]$ correspond to the robot's velocity and steering angle, respectively.

The constraints on control velocity v_i and control angle β_i are defined as

$$v_i \in [v_i^{b\max}, v_i^{f\max}] \tag{6}$$

$$\beta_i \in [\beta_i^{\min}, \beta_i^{\max}] \tag{7}$$

where $v_i^{b\max}$ and $v_i^{f\max}$ represent the robot i's maximum speeds when moving backward and forward. The $\beta_i^{\max}$ and $\beta_i^{\min}$ denote the maximum and minimum steering angles.

C. Conflict-Based Search

CBS utilizes a two-level search strategy known as high- and low-level search. The high-level search handles collision avoidance between agents, while the low-level search computes individual agent paths that adhere to constraints set by the high-level. After the low-level search generates paths, the high-level search verifies these paths for any collisions. Upon detecting a collision (or conflict), the high-level search subdivides the search space into two subspaces. Each subspace restricts the use of the conflicting location for one of the agents involved in the collision. This division process employs a binary tree structure called a conflict tree, where each node specifies spatio-temporal constraints for the agents. For instance, if agents a_1 and a_2 collide at space ${s_i}^t$ at timestep t_c, one node in the conflict tree will prevent agent a_1 from occupying space ${s_i}^t$ at timestep t_c. In contrast, another node will impose the same restriction on agent a_2. This iterative process continues until collision-free paths are identified [31].

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Algorithm 1 Upper-Level Conflict Resolution Algorithm

Require: Mission objective, robot position, map information.
Ensure: Collision-free paths for multiple 4WIDS robots.

Initialize binary body conflict tree *OPEN*, root node root;
$root.constraints \leftarrow \emptyset$;
$root.solution \leftarrow$ ***Lower-level_path_search***(a_i), $\forall a_i \in \mathcal{A}$;
$root.cost \leftarrow$ ***Calculate_cost***$(root.solution)$;
$OPEN \leftarrow \{root\}$;
while $OPEN \neq \emptyset$ **do**
  $N^* \leftarrow min_{cost} N$, $\forall N \in OPEN$;
  $C \leftarrow$ ***Detect_collision***$(N^*.solution)$;
  if $C = \emptyset$ **then**
    return $N^*.solution$;
  else
    for $a_i \in C$ **do**
      $a_i.cons \leftarrow \langle a_i, \mathcal{R}(s_j{}^t), t - \Delta t, t + \Delta t \rangle$;
      $N^{*'} \leftarrow N^*$;
      $N^{*'}.constraints \leftarrow N^*.constraints \cup \{a_i.cons\}$;
      $N^{*'}.solution \leftarrow$ ***Lower-level_path_search***(a_i);
      if $N^{*'}.solution \neq \emptyset$ **then**
        $OPEN \leftarrow OPEN \cup \{N^{*'}\} \setminus \{N^*\}$;
      end if
    end for
  end if
end while
return $\emptyset$;

Algorithm 2 Lower-Level Path Search Algorithm

Require: Mission objective, robot position, map information, Motion primitive libraries.
Ensure: Single-robot planning path.

Initialize open list *OPEN*, root node *root*;
$root.state \leftarrow s_i{}^0$;
$root.time \leftarrow 0$;
$OPEN \leftarrow \{root\}$;
while $OPEN \neq \emptyset$ **do**
  $n^* \leftarrow \text{argmin} n.\mathcal{F}$, $\forall n \in OPEN$;
  if ***Near_goal***$(n^*.state, g_i)$ **then**
    $p_i' \leftarrow$ ***Reeds_shepp***$(n^*.state, g_i)$;
    if ***Check_collision***$(n^*.state, n.t, p_i')$ **then**
      $\mathcal{P}_i \leftarrow$ ***Backtracking_path***(n^*, p_i');
      return $\mathcal{P}_i$;
    end if
  end if
  $states \leftarrow$ ***Get_child_node***$(n^*.\text{state})$;
  for each $s \in states$ **do**
    $n^{*'}.state \leftarrow s$;
    $n^{*'}.t \leftarrow n^*.t + 1$;
    if ***Check_collision***$(n^{*'}.state, n^{*'}.t) \neq \emptyset$ **then**
      Calculate $n^{*'}.\mathcal{H}$, $n^{*'}.\mathcal{G}$, $n^{*'}.\mathcal{F}$ using Eq. (8)-(11);
      if $n^{*'} \notin OPEN$ **then**
        $OPEN \leftarrow OPEN \cup \{n^{*'}\}$;
      else if $n^{*'}.\mathcal{G} < n^*_{OPEN}.\mathcal{G}$ **then**
        Update n^*_{OPEN} with $n^{*'}.state$, $n^{*'}.\mathcal{F}$;
      end if
    end if
  end for
end while
return $\emptyset$;

IV. Methodology

This section introduces a MAPF framework designed for 4WIDS robots, as illustrated in Fig. 1. The upper level extends CBS to address physical collisions among nonideal particle robots effectively. In contrast, the lower level integrates an enhanced spatio-temporal A* algorithm that considers the motion constraints of the robots. Additionally, a suboptimal accelerated search mechanism is incorporated to reduce computation time while preserving solution quality significantly.

A. Upper-Level Conflict Resolution

In the upper-level search, instead of traditional conflict detection using vertex, edge, or swap conflicts, we introduce a binary body conflict tree designed for continuous spaces. This constraint tree contains nodes, each representing a set of constraints, a solution, and its corresponding cost. The pseudocode is provided in Algorithm 1.

Initially, the root node of the body conflict tree is empty. The lower-level path planner *Lower-level_path_search* then generates a collision-free shortest path for each robot that satisfies kinematic constraints, forming the solution N.solution. Next, conflict detection is performed between pairs of robots using the body shape function *Detect_collision*. If no conflicts are found, the solution is feasible. If conflicts are detected, a spatio-temporal constraint $\langle a_i, \mathcal{R}(s_j{}^t), t - \Delta t, t + \Delta t \rangle$ is added for one of the robots, preventing it from reaching $s_j{}^t$ within the specified time window. The child nodes are then expanded, and the lower-level search is repeated until no conflicts remain or no solution is found.

B. Lower-Level Path Search

We propose an enhanced spatio-temporal hybrid A* algorithm for single-robot pathfinding. This approach integrates the hybrid A* method with the kinematic characteristics of multimodal 4WIDS robots, replacing grid-based nodes with motion primitives as search states. It also incorporates the cost of the motion mode switch during pathfinding. The optimal path and its associated cost are then passed to the upper level for conflict resolution. The pseudocode is presented in Algorithm 2.

Compared to traditional hybrid A*, our method enhances child node expansion and heuristic design by accounting for the multimodal motion characteristics of 4WIDS robots. When expanding child nodes using the *Get_child_node* function, the robot's motion is discretized into motion primitive libraries, with penalties assigned to each motion mode. The $\mathcal{G}$ function incorporates penalties for steering and mode switching. The heuristic function $\mathcal{H}$ combines the incomplete constraint heuristic cost in obstacle-free environments with the complete constraint heuristic cost in the presence of

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obstacles. It evaluates the Reeds–Shepp curve length at the goal, the Manhattan distance to the target, and the shortest path cost from the current node to the goal based on traditional A*. Additionally, to minimize frequent mode switching and optimize the driving path, $\mathcal{H}$ considers the difference between the current position and target orientation, further improving planning performance. The search formula is as follows:

$$\mathcal{F}(s_i{}^c, t_c, s_i{}^t, g_i) = \mathcal{G}(s_i{}^c, t_c, s_i{}^t) + \mathcal{H}(s_i{}^c, g_i) \tag{8}$$

$$\mathcal{G}(s_i{}^c, t_c, s_i{}^t) = \mathcal{G}(s_i{}^c, t_c, s_i{}^{t-1}) + \mathcal{C}(s_i{}^c, t_c) \tag{9}$$

$$\mathcal{C}(s_i{}^c, t_c) = \alpha \cdot f_1(t_c) + \beta \cdot f_2(t_c) + \gamma \cdot f_3(t_c) + \delta \cdot f_4(t_c) \tag{10}$$

$$\mathcal{H}(s_i^c, g_i) = \max\left(h_{\mathrm{rs}}(s_i^c, g_i), h_{\mathrm{o}}(s_i^c, g_i), h_{\mathrm{A}}(s_i^c, g_i)\right) + h_{\mathrm{h}}(s_i^c, g_i) \tag{11}$$

where $s_i{}^c$ is the current path node, t_c is the current time, and $s_i{}^t$ denotes the ith path node in the node set. $\mathcal{H}(s_i{}^c, g_i)$ represents the heuristic cost from the current node to the target node g_i. $\mathcal{C}(s_i{}^c, t_c)$ is the state-switching cost factor, with values based on straight movement, rotation, or waiting. The parameters $\alpha, \beta, \gamma, \delta$ correspond to steering, direction change, rotation, and lateral penalties, respectively. f_1, f_2, f_3, f_4 indicate the preceding status flags. $h_{\mathrm{rs}}(s_i^c, g_i)$, $h_{\mathrm{o}}(s_i^c, g_i)$, and $h_{\mathrm{A}}(s_i^c, g_i)$ represent the length cost of the Reeds–Shepp curve, the Manhattan distance, and the heuristic cost from the traditional A* algorithm, respectively, from the current node to the goal, and $h_{\mathrm{h}}(s_i^c, g_i)$ represents the orientation cost.

Based on this, the algorithm uses the function *Near_goal* to determine if the robot is near the target. Once triggered, it calculates the shortest Reeds–Shepp path from the current node to the goal using the *Reeds–shepp* function, considering the robot's motion state. If the robot is in Ackermann mode, the Reeds–Shepp curve for that state is calculated; in the lateral motion state, the corresponding lateral Reeds–Shepp curve is computed. The *Check_collision* function is then used to perform static obstacle detection along the path. If no collisions with environmental obstacles are detected, the path is deemed feasible, and the search terminates. The path is then obtained through backtracking using the *Backtracking_path* function. It is important to note that performing this Reeds–Shepp path check for every node may be resource-intensive. Therefore, after approaching the target point, the checks are performed at node intervals, with their frequency increasing as the robot approaches the goal.

C. Suboptimal Solution Acceleration

When the number of robots is small, optimal solvers perform efficiently. However, as the number of robots increases, the state space expands exponentially, rendering the problem NP-hard. The presence of multiple motion modes in 4WIDS robots further exacerbates this growth, making optimal solutions computationally infeasible for large-scale scenarios. In contrast, suboptimal solvers, with their faster computational efficiency, are a preferred choice when handling a large number of robots.

Here, the idea of balancing optimality and runtime is introduced using the parameter ω through a focused search approach. Under the hierarchical search framework, both the upper and lower levels maintain an OPEN list and a FOCAL list. The FOCAL list contains nodes from the OPEN list that meet the requirement, where the $f(n)$ value of a node must be within ω times the optimal node's $f_{\min}(n)$. Specifically, nodes in the FOCAL list are sorted based on their computed cost to prioritize their processing. The FOCAL list is represented as

$$FOCAL = \{n \in OPEN \mid f(n) \leq \omega \cdot f_{\min}(n)\} \tag{12}$$

where n denotes a path node, and the OPEN list contains all potential nodes, including the current node, in both the upper and lower level algorithms. The cost estimate of node n from the start to the goal is represented by $f(n)$, while $f_{\min}(n)$ denotes the minimum cost estimate within the OPEN lists.

To further accelerate the search, the improved temporal A* algorithm in the lower level introduces an adaptive heuristic weight factor (AHWF) ξ. This factor biases the search toward the goal and dynamically adjusts based on the robot's current position and distance to the target, balancing optimality and computational speed. The improved formula is as follows:

$$\mathcal{F}(s_i{}^c, t_c, s_i{}^t, g_i) = \mathcal{G}(s_i{}^c, t_c, s_i{}^t) + \xi \cdot \mathcal{H}(s_i{}^c, g_i). \tag{13}$$

V. Experimental Result

To verify the proposed method, sensitivity analysis and comparison experiments were conducted based on the designed scenarios and evaluation indicators. In addition, the ablation study further confirmed the effectiveness of the method.

A. Implementation Details

To thoroughly validate the reliability of the proposed method, as shown in Fig. 3, benchmark and application scenarios were designed. The benchmark scenario follows the continuous workspace reference proposed in [17], containing various numbers of robots across maps of sizes 50×50 and 100×100, with and without obstacles. The application scenarios include a warehouse with systematically arranged shelves and a factory inspection environment with dense equipment and cluttered corridors, resulting in more constrained and complex spaces. Although the map size is consistent with the benchmark scenario, the reduced navigable area necessitated fewer robots during testing.

Based on the benchmark tests, all robots are assumed to be homogeneous, with the following parameters: the robot's dimensions are 2 m $\times$ 3 m, with a front length $L_f = 2$m, rear length $L_b = 1$m, maximum forward and backward speed $v_{\max} = 2m/s$, and a minimum turning radius $r = 3$m. Notably, all programs are executed on a PC running Ubuntu 20.04, equipped with an Intel Core i7-10750H processor, Radeon Graphics at 2.60 GHz, and 16 GB of RAM.

B. Parametric Sensitivity Analysis

To determine the parameter choices for the proposed method, we first conducted sensitivity experiments. A typical scenario with 30 robots on a 100 $\times$ 100 map without obstacles was selected to test different parameters with 0.1 intervals from 1 to 2: the focused search parameter ω and the heuristic

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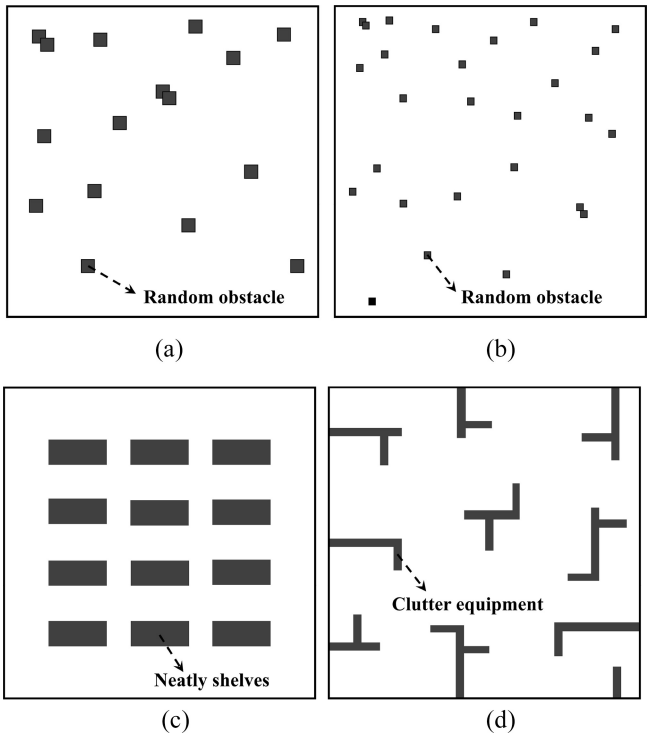


Fig. 3. Experimental scenarios. (a) Small benchmark scenario. (b) Large benchmark scenario. (c) Warehouse scenario. (d) Factory scenario.

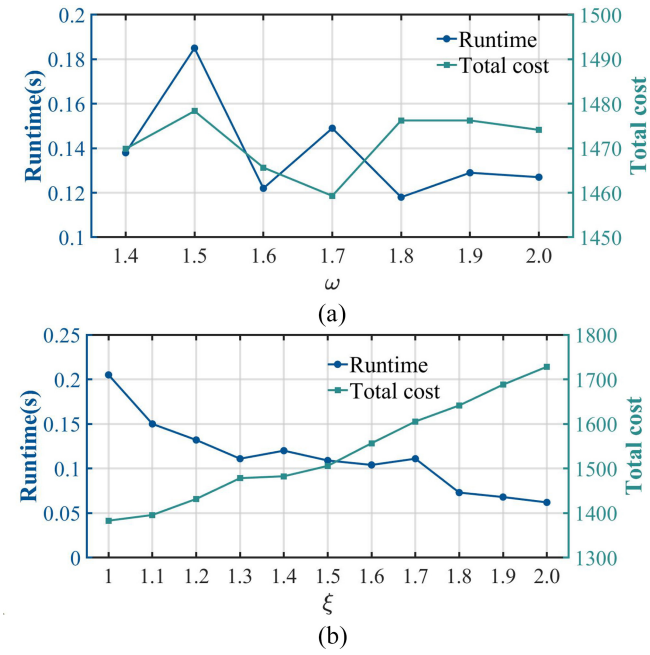


Fig. 4. Experimental results of parametric sensitivity analysis. (a) Experimental results of different focused search parameters ω. (b) Experimental results of different heuristic weight factors ξ.

weight factor ξ. Notably, we set the upper and lower focused search parameters to be identical and plotted the runtime and total cost for different parameters. The result is shown in Fig. 4.

The experimental results reveal that when the focused search factor is below 1.4, the algorithm struggles to find a solution due to scene complexity and extensive state expansion. As the focused search factor increases, runtime initially fluctuates but eventually decreases while the total cost remains relatively stable. At a focused search factor of 1.8, the total cost increases by 1.2% compared to the minimum, yet solution speed improves by 20.8%, indicating that a higher focused search factor can accelerate computation with minimal cost impact. However, when the focused search factor is 1.8, and a heuristic weight factor is introduced, the runtime is markedly reduced at the expense of a nearly linear increase in total cost. Specifically, with a heuristic weight factor of 2.0, the total cost increases by 25.0% while solution speed improves by 69.8%. These findings suggest that when confronted with large, complex scenarios, incorporating a focused search factor can accelerate the solution process. Additionally, starting with a high-heuristic weight to quickly approach the goaland then gradually reducing it helps minimize costs while balancing runtime and efficiency.

C. Comparative Experiments

As emerging robotic platforms, 4WIDS robots with kinematic constraints have received little research attention. Therefore, this article employs six advanced baseline algorithms for car-like robots CL-CBS [17], ICBS-H [32], ICL-CBS [33], CSDO [34], PSB [20], and HGKC [35] that incorporate vehicle kinematics. Notably, the PSB algorithm uses Bézier control points for trajectory optimization but lacks explicit waypoint specification, precluding path derivation and limiting quantification to partial metrics. Additionally, as [35]'s source code was unavailable, we improved our code using their paper's ideas to develop HGKC. Experiments conducted in both benchmark and application scenarios yield results summarized in Tables I and II.

Experimental results demonstrate that the proposed method significantly outperforms existing approaches in benchmark scenarios across runtime, execution time, travel distance, and total cost. While suboptimal solutions were introduced to enhance computational efficiency, the multimodal motion capabilities of 4WIDS robots ensure superior overall performance. Notably, compared to the CL-CBS, ICBS-H, ICL-CBS, CSDO, PSB, and HGKC algorithms, the runtime of this method in the four experimental scenarios is reduced by 73.3%, 68.5%, 49.6%, 19.7%, 91.0%, and 52.9% on average, and the total cost is reduced by 7.1%, 7.2%, 8.5%, 6.2%, 0.5%, and 4.5% on average.

In the application scenario, the proposed method outperforms alternative approaches in runtime, execution time, and travel distance. Although the other methods reach the target slightly faster in simple environments, the proposed method outperforms it in complex scenes, with its runtime advantage becoming increasingly pronounced as complexity grows. However, as the number of robots and scenario complexity increase, the benefit in total cost diminishes. This reduction is due to the narrow, intricate channels in the application scenarios that require frequent mode switching, thereby increasing total cost. Importantly, the increased execution cost does not compromise the optimality of execution time or travel distance.

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TABLE I
EXPERIMENTAL RESULTS OF DIFFERENT PLANNING ALGORITHMS IN BASELINE SCENARIOS

Map size	Robots	Method	Without obstacle				With obstacle			
			Runtime	Average execution time	Average travel distance	Total cost	Runtime	Average execution time	Average travel distance	Total cost
50×50	15	CL_CBS	0.15	14.07	25.62	488.20	1.96	14.27	25.91	516.72
		ICBS-H	0.14	14.07	25.66	488.72	1.18	14.60	25.55	517.89
		ICL_CBS	0.07	14.27	26.12	495.61	0.12	14.13	25.60	539.66
		CSDO	0.07	15.74	28.77	503.37	**0.03**	13.01	23.21	402.16
		PSB	0.66	—	—	462.94	0.90	—	—	**379.25**
		HGKC	0.05	13.47	**23.39**	**455.09**	0.41	14.27	24.23	499.31
		Ours	**0.02**	**13.33**	23.77	463.84	0.05	**12.87**	**22.57**	434.19
100×100	30	CL_CBS	0.27	23.83	45.92	1559.40	0.93	23.57	45.66	1567.98
		ICBS-H	0.24	23.83	45.92	1559.40	0.71	23.60	45.59	1565.99
		ICL_CBS	0.25	24.07	46.26	1577.02	0.41	23.68	45.82	1576.43
		CSDO	0.25	26.08	52.11	1682.73	0.45	27.43	53.90	1757.86
		PSB	0.98	—	—	1618.80	2.56	—	—	1663.76
		HGKC	0.22	23.52	44.72	1549.05	0.44	23.38	44.36	1568.88
		Ours	**0.17**	**22.47**	**43.55**	**1501.62**	**0.26**	**22.39**	**43.26**	**1508.21**

TABLE II
EXPERIMENTAL RESULTS OF DIFFERENT PLANNING ALGORITHMS IN APPLICATION SCENARIOS

Scenario	Robots	Method	Runtime	Average execution time	Average travel distance	Total cost
Warehouse 50×50	5	CL_CBS	0.28	15.60	29.03	175.86
		ICBS-H	0.18	15.60	29.03	177.98
		ICL_CBS	0.09	15.80	29.27	182.36
		CSDO	**0.01**	17.00	32.38	**145.39**
		PSB	0.37	-	-	172.36
		HGKC	0.29	**15.20**	**26.48**	172.84
		Ours	0.09	16.25	28.93	146.14
Warehouse 100×100	8	CL_CBS	0.42	26.25	51.41	548.94
		ICBS-H	0.41	26.25	51.41	**458.94**
		ICL_CBS	**0.31**	26.25	51.27	463.09
		CSDO	0.37	29.00	56.95	503.27
		PSB	0.75	-	-	491.94
		HGKC	0.45	29.25	56.76	518.34
		Ours	**0.31**	**25.88**	**50.32**	472.31
Factory 50×50	5	CL_CBS	0.09	**14.80**	28.04	186.79
		ICBS-H	0.08	**14.80**	28.04	186.80
		ICL_CBS	0.15	16.40	30.77	194.07
		CSDO	**0.01**	15.00	27.84	168.82
		PSB	0.30	-	-	178.77
		HGKC	0.10	15.00	27.02	179.08
		Ours	0.02	15.40	**26.71**	**154.61**
Factory 100×100	8	CL_CBS	0.58	29.25	57.39	**508.88**
		ICBS-H	0.55	29.25	57.39	**508.87**
		ICL_CBS	0.39	29.00	57.06	509.40
		CSDO	0.35	**27.63**	**53.76**	558.69
		PSB	0.88	-	-	572.31
		HGKC	0.49	29.00	56.76	509.40
		Ours	**0.23**	29.00	57.26	557.38

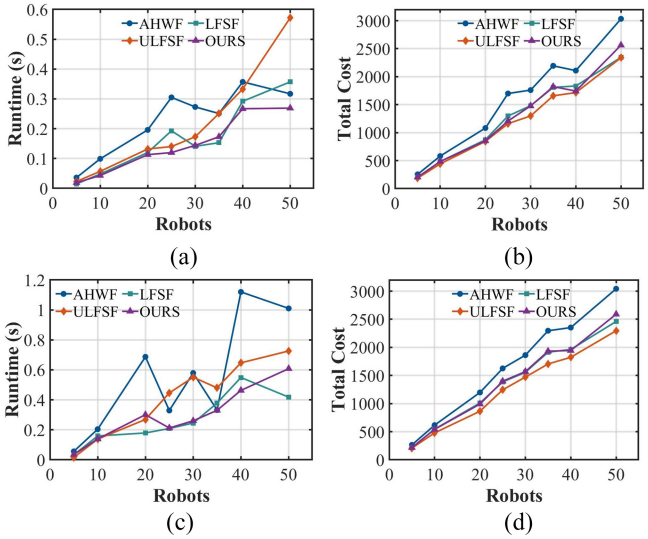


Fig. 5. Experimental results of ablation experiment. (a) Experimental results of runtime without obstacle scenarios. (b) Experimental results of total cost without obstacle scenarios. (c) Experimental results of runtime with obstacle scenarios. (d) Experimental results of total cost with obstacle scenarios.

D. Ablation Experiment

To assess the impact of each component in the proposed method, ablation experiments were conducted on a 100 × 100 map. The evaluation involved varying robot numbers across four methods: 1) AHWF; 2) Lower-level focused search factor (LFSF); 3) upper and lower level focused search factor (ULFSF); and 4) the full proposed method. The runtime and total cost for each algorithm under different robot counts are depicted in Fig. 5.

The results indicate that the proposed method achieves near-optimal runtime performance. While its total cost is slightly higher than that of the ULFSF due to the incorporation of adaptive heuristic weights, it significantly improves runtime efficiency. Among the tested algorithms, the LFSF ranks second in runtime performance, followed by ULFSF. The additional upper-level search in ULFSF increases computational overhead, leading to longer runtimes than LFSF. However, ULFSF compensates for this by achieving a lower total cost, prioritizing cost efficiency at the expense of runtime. Although adaptive heuristic weights enhance runtime performance, their impact is relatively moderate. When integrated with ULFSF,

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the proposed method effectively balances optimality and efficiency, demonstrating superior overall performance. An advantage that becomes more pronounced as the number of robots increases and scenario complexity grows.

VI. Conclusion

This study addresses the MAPF problem for 4WIDS robots by proposing an enhanced CBS framework that integrates multiple motion modalities. In the upper level, conflicts between robots are resolved by considering physical collisions, while the lower level optimizes single-robot path planning under vehicle kinematics constraints. Additionally, a suboptimal solution search acceleration strategy is introduced to balance solution quality and computational efficiency. Extensive experiments demonstrate that our approach significantly outperforms advanced CL-CBS, ICBS-H, ICL-CBS, CSDO, PSB, and HGKC algorithms reducing runtime by 73.3%, 68.5%, 49.6%, 19.7%, 91.0%, and 52.9% and total cost by 7.1%, 7.2%, 8.5%, 6.2%, 0.5%, and 4.5% on average across four benchmark scenarios. Notably, the advantages of our method become more pronounced as the number of robots and scenario complexity increase, effectively balancing optimality and efficiency. Overall, this study provides valuable insights into MAPF and advances the practical deployment of 4WIDS robots.

Furthermore, this study assumes a constant agent velocity rather than explicitly planning it. For multiagent collision detection and avoidance, in addition to imposing constraints to prevent spatial conflicts, velocity control can be employed to ensure robots reach potential collision points at different times, effectively mitigating collisions. This aspect will be a key focus of future research.

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